

MARINA DI GINOSA, Italy

WMO: 163250

Lat: 40.424N

Lon: 16.887E

Elev: 38

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.9	36.2	24.9	18.9	46.7	28.4	22.4	44.6	33.0	56.0	30.0	56.1	10.4	290	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	94.6	74.5	91.5	74.9	89.3	74.6	82.9	87.8	81.0	86.5	79.2	85.1	10.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.6	164.6	87.1	79.2	152.0	85.4	77.3	142.2	83.8	47.0	87.6	44.8	86.7	42.8	85.6	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.7	22.1	18.7	DB	30.3	100.2	2.2	4.0	28.8	103.1	27.5	105.5	26.3	107.7	24.8	110.7	
			WB	28.8	85.8	2.2	3.4	27.3	88.3	26.0	90.3	24.8	92.2	23.2	94.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	48.8	49.5	53.3	58.0	66.2	74.2	79.7	79.6	72.4	65.0	57.1	50.9
	DBStd	12.05	4.86	4.96	4.78	4.61	4.62	5.11	4.11	4.13	4.73	4.86	5.55	5.16
	HDD50	225	78	62	23	4	0	0	0	0	0	0	10	48
	HDD65	2289	501	434	362	212	39	2	0	0	5	58	240	436
	CDD50	4972	43	48	126	246	504	725	922	918	671	467	225	77
	CDD65	1557	0	0	0	3	78	277	457	453	226	59	4	0
	CDH74	14607	0	0	3	18	442	2402	5086	4897	1592	163	4	0
	CDH80	5529	0	0	0	1	76	801	2217	2068	358	8	0	0
Wind	WSAvg	8.9	9.4	9.9	9.9	9.0	8.6	8.3	8.5	8.2	8.4	8.5	9.1	9.0
Precipitation	PrecAvg	17.7	2.0	1.7	1.7	1.2	1.1	0.8	0.7	0.8	1.7	2.2	2.3	2.3
	PrecMax	28.3	8.3	4.9	5.1	6.1	3.3	3.1	3.0	4.4	7.2	7.8	8.1	9.4
	PrecMin	6.7	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.9	1.9	1.3	1.2	1.1	0.8	0.7	0.7	0.9	1.7	1.9	1.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.7	65.5	71.5	76.1	86.0	95.4	98.8	98.6	89.8	80.9	73.4	66.4
		MCWB	56.6	56.0	55.5	61.9	68.3	72.6	71.7	75.0	73.0	69.8	67.1	60.5
	2%	DB	61.0	62.4	67.1	72.0	82.1	90.2	94.8	94.6	86.3	77.4	70.2	63.6
		MCWB	56.4	55.9	56.1	60.0	67.3	71.1	74.4	75.4	74.2	69.4	64.8	58.7
	5%	DB	59.1	60.5	64.1	69.6	78.8	87.6	91.7	91.3	84.1	75.5	68.3	61.9
		MCWB	55.3	55.2	56.2	59.4	66.9	72.1	74.6	76.3	74.4	68.5	63.3	57.4
	10%	DB	57.4	58.8	61.4	66.5	75.6	84.5	89.6	88.7	81.8	73.6	66.4	59.4
		MCWB	53.7	54.5	55.7	59.2	66.3	72.0	74.9	76.1	73.4	67.7	62.2	55.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	60.3	61.3	66.6	74.7	81.9	85.4	85.5	81.1	74.7	68.8	62.2
		MCDB	62.2	62.4	64.9	71.8	79.7	86.1	90.3	90.2	84.8	77.7	71.5	65.0
	2%	WB	57.8	58.5	59.6	63.8	72.0	78.9	82.9	83.4	78.9	72.2	66.6	59.8
		MCDB	60.2	60.6	63.2	68.6	77.5	84.3	88.0	88.3	82.9	75.3	69.0	62.7
	5%	WB	56.1	56.8	58.1	62.1	70.0	76.7	80.9	81.4	76.9	70.4	64.4	57.8
		MCDB	58.6	59.5	61.9	66.4	75.5	83.1	86.9	86.8	81.6	73.7	67.5	60.7
	10%	WB	54.2	54.8	56.7	60.5	68.1	75.0	78.9	79.6	74.7	68.7	62.6	56.1
		MCDB	57.2	57.9	60.6	64.6	73.4	81.2	85.6	85.4	79.6	72.5	66.2	59.1
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.4	14.8	15.4	15.9	16.8	17.7	17.3	16.0	14.7	13.6	13.2
		MCDBR	15.2	17.0	19.0	19.8	20.1	20.5	21.0	20.0	17.8	16.5	14.8	14.3
		MCWBR	11.7	12.3	12.2	12.0	11.8	12.1	13.1	12.7	12.1	11.3	11.3	11.1
	5% WB	MCDBR	13.3	14.4	15.6	16.7	17.1	17.2	17.7	17.0	15.7	14.2	12.1	12.6
		MCWBR	11.7	12.4	12.3	12.6	12.8	13.3	14.8	14.4	13.3	11.7	10.7	10.9
Clear-Sky Solar Irradiance	taub	0.343	0.366	0.405	0.453	0.449	0.435	0.430	0.438	0.432	0.401	0.371	0.347	
	taud	2.447	2.358	2.248	2.120	2.195	2.280	2.299	2.281	2.293	2.364	2.398	2.437	
	Ebn at Noon	257	266	267	261	265	268	268	262	256	251	243	245	
	Edh at Noon	25	32	40	48	46	42	41	41	37	31	27	24	
All-Sky Solar Radiation	RadAvg	599	839	1242	1616	1969	2227	2296	2023	1478	1038	679	538	
	RadStd	62	72	126	128	117	120	100	121	83	84	58	63	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-110	+164	N/A

Nomenclature: See separate page